GPU ALGORITHMS / EXPERIMENTAL STUDY

Stable GPU radix sorting with recycled pages

An experimental radix sort that keeps the result in the original array and uses a small reusable workspace.

STABILITY REQUIRED

14 SEPTEMBER 2026 • CCCL 3.6.0 CHECKOUT

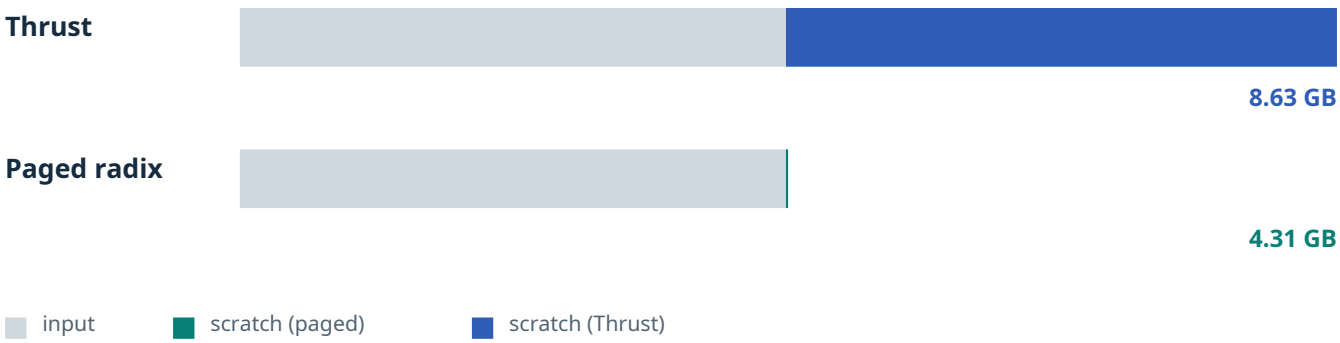
16.88 MB
temporary storage

60.6%
Thrust throughput retained

4.31 GB
accounted peak memory

Measured for **1,073,741,824 random 32-bit keys** on one RTX PRO 6000 Blackwell. The comparison is against the CCCL 3.6.0 checkout through Thrust's stable radix-sort path.

The memory difference



Peak = input + measured scratch. CUDA runtime/context and benchmark-only copies are excluded. Bars use a common linear scale; the 16.88 MB paged workspace is a thin sliver beside the 4.29 GB input. Timings use reused allocations; input resets and allocation growth are excluded.

CUDA 12.9 / 9 TRIAL MEDIANS / 5 WARMUPS / REUSED ALLOCATIONS